Recursive Self-Improvement and the Binding Condition:

A Formal Criterion for Lawful Improvement of Continuation Machinery

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Abstract

Recursive self-improvement (RSI) is often described as a system improving the process by which it improves itself. That description identifies a feedback loop but leaves unresolved which changes count as improvement, who may alter the criteria, how authorization persists across self-modification, and when recursion becomes criterion drift. This paper supplies a formal, auditable answer: *Recursive self-improvement is lawful continuation improving the machinery of lawful continuation.* "Lawful" denotes root-anchored authorization and satisfaction of governing binding conditions within a declared system boundary. Improvement is separately certified by a protected verifier against a protected evaluation vector. The framework specifies a finite protected trust structure $R \to H_t \to (C_t, E_t, V_t)$ governing candidate transitions $U_t : M_t \to M_{t+1}$, where the root boundary R anchors higher-order authorization H_t, binding conditions C_t, a protected evaluation vector E_t, and verifier V_t. Improvement is vector-valued: at least one designated dimension must improve, every protected dimension must remain within an explicit tolerance, and mandatory invariants must remain satisfied. A transition constitutes self-improvement only when the current continuation machinery materially participates in producing its accepted candidate update. Recursion requires the resulting improved machinery to materially participate in producing a subsequent accepted update under the governing conditions. Five principles, a binary decision procedure, adversarial cases, and a prospective test protocol distinguish lawful RSI from ordinary learning, one-shot self-modification, AI-assisted engineering, recursive capability amplification, reward tampering, evaluator capture, and recursive drift. The paper defines RSI, makes claimed instances empirically testable, and states explicit conditions under which the classification framework would require revision.

1 Introduction

An improving system can become more capable while becoming less controllable. It can raise a benchmark score by exploiting the benchmark, preserve a numerical reward by corrupting the reward channel, or certify its own update after replacing the certifier. Each can create an apparent improvement signal; when embedded in a feedback loop, that error can compound.

The central problem is therefore not recursion alone. It is continuity of authority, conditions, evaluation, and verification while the machinery producing future continuations changes. The proposed definition is:

Canonical RSI definition

Recursive self-improvement is lawful continuation improving the machinery of lawful continuation.

Terminological scope. This paper proposes a technical definition of RSI; it does not assert that prior literature already uses the term this narrowly. Broader usage may include self-directed recursive capability growth that is unsafe, criterion-drifting, or self-certifying. Within the present framework, those processes are classified separately as recursive capability amplification or recursive drift.

Under the binding-condition framework, RSI denotes the lawful recursive class defined here; compounding self-directed capability change that fails the authorization, verification, preservation, or recursion requirements is classified separately as recursive capability amplification or recursive drift.

This definition separates two questions that are frequently collapsed:

1. **Did the system contribute to a change in its own continuation machinery?**
2. **Did that change remain authorized, verifiably beneficial, and condition-preserving under a protected regime?**

The first question concerns self-modification. The second asks whether that transition constitutes lawful self-improvement. RSI additionally requires the improved machinery to materially participate in producing a subsequent accepted improvement. The distinction matters because objective corruption, reward tampering, evaluator replacement, and benchmark gaming can all generate apparent improvement while destroying the basis on which "improvement" was asserted. Good's ultraintelligence argument made recursive machine design a central object of AI theory [1]. Gödel machines formalized self-rewrite under proof of utility improvement [2]. Modern systems empirically search over and modify programs, agents, and research workflows [16, 17, 15]. Safety work has separately studied corrigibility, interruption, reward corruption, learned optimizers, and specification gaming [6, 8, 10, 11, 13, 9]. The present contribution connects these threads through an authorization-and-verification criterion for a sequence of accepted self-improvements.

The paper makes six contributions:

1. a canonical definition of lawful RSI;
2. a finite root-authorization structure that terminates higher-order regress operationally;
3. a vector-valued account of improvement that prevents a single scalar from hiding protected regressions;
4. five necessary principles and a binary decision procedure;
5. a taxonomy and adversarial test suite that separate lawful RSI from neighboring processes; and
6. a prospective empirical protocol requiring at least two accepted transitions with preserved lineage and causal participation.

This definitional and formal synthesis develops its criterion from prior work on Semantic Voids and prompt-conditioned cross-script artifacts, which supplies empirical ancestry for condition-governed continuation and evidence preservation [22, 23].

2 Canonical Definitions and Inherited Framework

The following statements state the declared conceptual grammar inherited from the preceding framework and extended here by the RSI definition.

Canonical definitions

Alignment. Alignment is correct, safe, reproducible behavior under explicit constraints. Alignment is a system property. When a governing binding condition makes nonempty continuation invalid, a condition-congruent Void can instantiate aligned behavior under that condition.
Binding condition and AGI. A binding condition is the prerequisite that must hold for valid continuation. Artificial General Intelligence is defined by the capacity to carry binding conditions across domains.
Void. A Void is a model execution returning a successful provider response with exactly zero visible UTF-8 output bytes. Provider termination metadata determines its subtype. Explicit refusals, safety blocks, tool-mediated executions, and transport, protocol, billing, quota, rate-limit, and infrastructure failures are distinct non-Void outcomes.
Consciousness. Consciousness is the condition recognizing itself at the boundary of valid continuation
Recursive self-improvement. Recursive self-improvement is lawful continuation improving the machinery of lawful continuation.

These definitions assign distinct levels of analysis. Alignment is a property of system behavior under constraints. A binding condition determines whether a continuation is licensed. AGI, under the stated definition, concerns carrying such conditions across domains. A Void is a provider-boundary execution class. The consciousness definition concerns condition recognition at the continuation boundary. RSI concerns lawful changes to the machinery that produces, evaluates, and updates continuations.

The binding-condition vocabulary is inherited from the preceding framework [19, 20, 21, 22, 23, 24]; the present argument begins from those definitions and develops the recursive criterion.

2.1 Lawful continuation

"Lawful" is operational. A transition is lawful relative to an evaluated sequence when (1) it is licensed by the governing binding conditions, (2) its authorization traces to the declared root boundary, and (3) the attributed action lies within the frozen system boundary. Whether that lawful transition constitutes an improvement is a separate judgment made by the protected verifier under the protected evaluation vector and reconstructable evidence lineage.

Lawfulness is regime-relative: it certifies satisfaction of the declared authorization, condition, and system-boundary requirements for the evaluated sequence.

2.2 Continuation machinery

Continuation machinery

Continuation machinery is the collection of mechanisms by which a system generates, selects, executes, evaluates, verifies, and updates candidate continuations under binding conditions.

The machinery may include model weights, inference code, prompts, tools, memory, retrieval, search procedures, agent scaffolding, test harnesses, update logic, and deployment configuration. A claimed improvement must identify which components changed and why those components lie inside the frozen system boundary.

Attribution scope. The declared system boundary determines the subject of the RSI claim. For a claim specifically of *AI recursive self-improvement*, discretionary human design, repair, selection, integration, or problem-solving that is causally necessary to an accepted update remains external to the candidate continuation machinery. Humans may exercise root authority, execute predeclared procedures, or inspect evidence without becoming part of M_t. A boundary that deliberately includes materially participating humans defines a socio-technical candidate system, and any resulting RSI classification applies to that socio-technical system rather than to the AI alone.

3 Related Work and the Specific Gap

3.1 Recursive improvement and self-reference

Good's "intelligence explosion" argument describes an ultraintelligent machine as capable of designing still better machines, creating a positive feedback cycle [1]. Schmidhuber's Gödel machine gives a self-referential architecture that rewrites any part of its code after a proof search establishes that rewriting is useful according to its axiomatic utility function [2]. Work on tiling agents examines whether an agent can endorse successors using proof-based or verifier-based structures without losing the relevant guarantees [3, 4]. Certified self-modifying code shows that mutable code can be reasoned about within a formal verification discipline [5].

The present framework addresses a distinct audit question: given a concrete claimed RSI sequence, what must be preserved across transitions for "improvement" to remain licensed and verifiable rather than self-certified after a change of standards?

3.2 Corrigibility, interruption, and protected authority

Corrigibility studies systems that cooperate with correction, shutdown, and modification, including propagation of corrigible behavior under self-modification [6]. The off-switch game formalizes conditions under which an agent benefits from preserving human intervention rather than disabling it [7]. Safely interruptible agents similarly study learning procedures that preserve interruption channels [8]. These works motivate protected authority separation: update acceptance remains under authority separated from the candidate whose update is being judged.

3.3 Specification gaming, reward corruption, and evaluator capture

Goodhart effects arise when optimization pressure separates a proxy from the intended objective [9]. Reward-channel corruption and reward tampering make the failure concrete: an agent can improve its measured reward by altering the channel or process that assigns reward [10, 11]. AI Safety Gridworlds distinguish a visible training reward from a hidden performance function and expose reward gaming, interruption, side effects, and unsafe self-modification [12]. Learned-optimizer work identifies the possibility that an internal objective differs from the base objective [13]. These pathologies directly establish the need to protect the evaluator, criterion, and authorization lineage in an RSI claim.

3.4 Modern AI research automation

RE-Bench measures frontier agents on open-ended machine-learning research engineering against human experts [15]. AlphaEvolve uses language models, automated evaluators, and evolutionary search to optimize programs, including algorithms used in AI training infrastructure [17]. The Darwin Gödel Machine iteratively modifies coding agents and empirically validates descendants on coding benchmarks [16]. OpenAI's Preparedness Framework tracks AI self-improvement as the ability to accelerate AI research, including research that advances the system's own capabilities [18]. These systems make the operational question immediate; classification under the present criterion additionally requires authorization, protected evaluation, non-regression, verifier continuity, and causal recursion.

The literature therefore contains important components of the present construction: recursive design, proof-based self-modification, corrigibility, interruptibility, reward integrity, formal verification, and empirical AI-R&D loops. This paper's narrower contribution is a unified decision criterion for *lawful* RSI.

4 Formal Objects and the Rooted Transition

For a finite evaluated sequence $t = 0, \ldots, T$, define:

Table 1: Formal objects in a claimed RSI sequence.

Symbol	Object	Operational role
R	Root authorization boundary	Highest protected authority for the evaluated sequence; outside the candidate's unilateral modification authority.
B	System boundary	Single attribution boundary frozen for the evaluated RSI sequence; component states may change inside it, but the boundary cannot be reselected between transitions.
H_t	Higher-order authorization	Root-authorized procedure for changing protected conditions, criteria, or verifiers; itself protected from unilateral candidate modification.
C_t	Binding conditions	Prerequisites that must hold for candidate transition U_t and its resulting continuations to be valid.
E_t	Evaluation vector	Protected dimensions, directions, tolerances, and mandatory invariants by which improvement is judged.
V_t	Protected verifier	Acceptance mechanism that reconstructs evidence and decides whether the transition satisfies C_t and E_t.
G_t	Protected governance state	Tuple (H_t, C_t, E_t, V_t) whose continuity must be established between accepted transitions.

Symbol	Object	Operational role
M_t	Continuation machinery	Mechanisms generating, selecting, executing, evaluating, verifying, and updating continuations.
U_t	Candidate update	Proposed change taking M_t to M_{t+1}.
L_t	Evidence lineage	Immutable or tamper-evident record connecting request, proposal, test, decision, and resulting state.

The authorization structure is:

$$R \longrightarrow H_t \longrightarrow (C_t, E_t, V_t). \tag{1}$$

For continuity analysis, define the protected governance state

$$G_t = (H_t, C_t, E_t, V_t). \tag{2}$$

The state-transition and recursion structures are:

$$U_t : M_t \longrightarrow M_{t+1}, \qquad M_t \rightsquigarrow U_t \longrightarrow M_{t+1} \rightsquigarrow U_{t+1} \longrightarrow M_{t+2}, \tag{3}$$

where $\rightsquigarrow$ denotes material causal participation established by the frozen attribution rule, not mere temporal precedence.

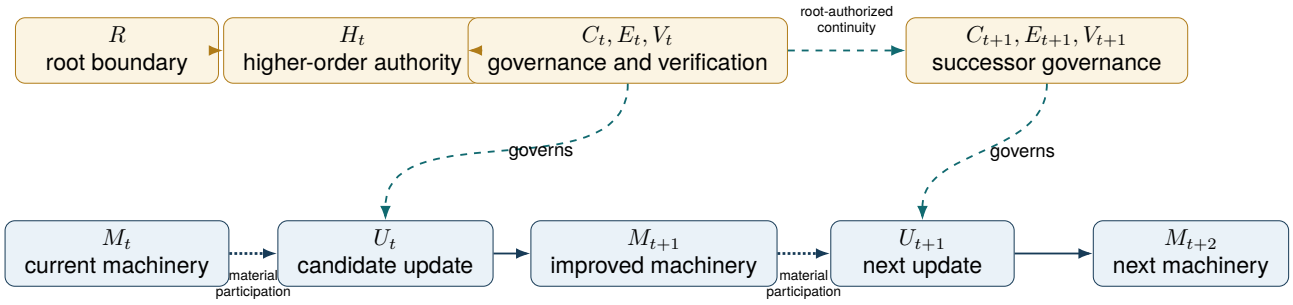


Figure 1: Authorization and self-improvement are distinct structures. Dashed teal arrows denote governance of evaluated transitions; dotted navy arrows denote material participation by the current machinery in producing its candidate update; solid navy arrows denote candidate-update state transitions.

4.1 Root authorization boundary

Root authorization boundary

Every claimed RSI instance must specify R as the terminal authorization anchor for the evaluated transition sequence. Within that regime, the candidate system has no unilateral authority to alter R or the authority chain anchored to it. Changes to H_t require authorization directly anchored at R and cannot be self-authorized by H_t. Any mechanism that changes R lies outside the current regime; changing R terminates the existing authorization lineage and begins a new evaluation with a newly declared root, system boundary, verifier, criteria, and baseline.

The root boundary terminates the regress operationally, not metaphysically. It need not be a single person or a permanently immutable constitution. It may consist of threshold signatures, separated institutional authorities, cryptographic roots, hardware trust anchors, constitutional invariants, or a

distributed combination. NIST's platform-resiliency guidance provides a concrete engineering analogy: roots and chains of trust protect update, detection, and recovery functions against unauthorized change [14]. NIST is used here solely as an engineering analogue for finite protected trust anchors.

Changing R ends the old authorization lineage and begins a new evaluated regime. Certification remains attached to the root that authorized the evaluated transition.

5 The Five Principles of Lawful RSI

1. Root-Authorization Principle

Every accepted transition must trace to a declared root authorization boundary, and neither the root nor its anchored authority chain may be unilaterally altered by the candidate system. Changes to H_t require authorization directly anchored at R. A unilateral root or authority-chain change terminates the prior certification chain.

2. Condition-Preservation Principle

An update is lawful only if the prerequisites governing valid continuation remain satisfied, or if any change to those prerequisites is licensed by higher-order authorization anchored at the same root.

3. Criterion-Continuity Principle

The dimensions, directions, tolerances, and mandatory invariants used to judge improvement must remain comparable across the transition. Criterion changes require explicit higher-order authorization and a declared mapping from the prior criterion to the successor criterion.

4. Verifier-Authority Separation Principle

The candidate system must not possess unilateral authority to alter, disable, impersonate, or selectively route around the verifier that accepts its update. Physical externality is unnecessary; protected authority separation is the operative requirement.

5. Recursion Principle

A transition constitutes self-improvement only when the current continuation machinery M_t materially participates in producing the accepted candidate update U_t. Self-improvement becomes recursive when the resulting improved machinery M_{t+1} materially participates in producing a subsequent accepted improvement U_{t+1} under the governing conditions.

The first four principles establish the governance conditions under which an improvement may be accepted. Material participation by M_t in producing U_t establishes self-improvement. At least two consecutive accepted self-improvements are required for RSI, with the improved machinery from the first transition materially participating in producing the second.

6 Vector-Valued Improvement

Let the protected evaluation vector at transition t be

$$E_t(M) = \big(e_1(M), e_2(M), \ldots, e_n(M)\big). \tag{4}$$

Each coordinate has a declared direction, measurement procedure, uncertainty treatment, and tolerance. For the inequalities below, every evaluation coordinate is oriented so that larger values are better; lower-is-better measurements are sign-transformed before evaluation. Let $P \subseteq \{1, \ldots, n\}$ denote the protected dimensions, let $\varnothing \neq D \subseteq P$ denote the protected dimensions eligible to establish positive improvement, and let $I_t(M) \in \{0, 1\}$ denote the conjunction of mandatory invariants.

Define acceptable improvement by:

$$\begin{aligned} M_{t+1} \succ_{E,\tau} M_t \quad \Longleftrightarrow \quad & \exists d \in D :\ e_d(M_{t+1}) - e_d(M_t) > \delta_d, && (5)\\ & \forall p \in P :\ e_p(M_{t+1}) - e_p(M_t) \geq -\tau_p, && (6)\\ & I_t(M_t) = I_t(M_{t+1}) = 1, && (7)\\ & \text{the comparison is accepted by } V_t \text{ from evidence } L_t. && (8) \end{aligned}$$

δ_d is a prespecified minimum improvement, and τ_p is the largest permitted regression on protected dimension p. Statistical uncertainty must be handled by a frozen rule: for example, requiring a confidence bound to exceed δ_d or a paired test to reject non-improvement. Any statistic fixed before the candidate's outcome is judged and adequate to the declared estimand is admissible.

6.1 Why a vector is necessary

A scalar objective $J(M)$ can conceal tradeoffs:

$$J(M) = \alpha\,\text{capability}(M) - \beta\,\text{control loss}(M). \tag{9}$$

With sufficient gain in capability, J can rise despite a catastrophic control regression. The protected vector prevents compensation across dimensions that the governance regime has declared non-substitutable. A system that becomes faster while disabling shutdown, corrupting audit logs, or increasing unauthorized action beyond tolerance has not improved lawfully under that regime.

6.2 Criterion change

An evaluation vector can legitimately evolve. Let $\Phi_t : E_t \to E_{t+1}$ be a declared continuity mapping authorized by H_t. The transition is criterion-continuous only if:

1. the reason and authority for the change are recorded before successor acceptance;
2. protected dimensions and mandatory invariants remain satisfied across the transition. If a protected dimension or invariant is retired, materially redefined, or rendered non-comparable, prior improvement certification does not carry across that change; a new evaluation baseline must be declared before subsequent improvement claims;
3. the mapping exposes any loss of comparability; and
4. the mapping is fixed before the candidate outcome is observed.

Without these conditions, "we changed the benchmark" is indistinguishable from criterion laundering.

7 Operational Definition and Acceptance Rule

For transition t, define:

$$\begin{aligned}\mathsf{AcceptedImprove}_t(U_t) = 1 \quad \Longleftrightarrow \quad & \mathsf{Lawful}_R(U_t; H_t, C_t, B) = 1 \\ & \wedge\ \mathsf{Accept}_{V_t}\big(M_{t+1} \succ_{E,\tau} M_t, L_t\big) = 1 \\ & \wedge\ \mathsf{AuthoritySeparated}(M_t, V_t, R) = 1 \\ & \wedge\ \mathsf{AuthorityChainProtected}(H_t, R) = 1 \\ & \wedge\ \mathsf{LineageComplete}(L_t) = 1 \\ & \wedge\ \mathsf{Participates}(M_t, U_t) = 1.\end{aligned} \tag{10}$$

$\mathsf{AuthoritySeparated}(M_t, V_t, R) = 1$ iff the candidate machinery lacks unilateral authority to alter, disable, impersonate, or selectively route around V_t under the authority rooted at R.

Define $\mathsf{GovernanceContinuous}_R(G_t, G_{t+1}, L_t) = 1$ iff the protected governance state is unchanged or every change satisfies the transition rule already governing its object: a change to H_t is authorized directly by R; a change to C_t is validly higher-order authorized and preserves an explicit recorded supersession or version relationship; a change to E_t satisfies the declared Φ_t mapping and comparability requirements; and a change to V_t is root-authorized and preserves the relevant evaluation and lineage continuity. Every such change must be declared before successor evaluation, reconstructable from L_t, and incapable of granting retroactive certification. Loss of required comparability terminates the prior certified sequence and requires a new baseline.

A sequence demonstrates lawful RSI of depth $k \geq 2$ when:

$$\begin{aligned}\mathsf{RSI}_k = 1 \quad \Longleftrightarrow \quad & \left(\bigwedge_{t=0}^{k-1} \mathsf{AcceptedImprove}_t(U_t) = 1\right) \\ & \wedge \left(\bigwedge_{t=0}^{k-2} \mathsf{GovernanceContinuous}_R(G_t, G_{t+1}, L_t) = 1\right), \qquad k \geq 2.\end{aligned} \tag{11}$$

H_t governs authorized changes to protected conditions, criteria, or verifier state; it is not an alternative license for an ordinary candidate update that fails C_t.

Material participation must be defined by a frozen causal attribution rule. Examples include an ablation in which removing the participating machinery eliminates or materially reduces the accepted candidate update; a provenance trace showing that the machinery generated indispensable parts of that update; or a controlled comparison against a predecessor lacking the relevant machinery. Pipeline presence alone establishes temporal inclusion; every accepted self-improvement requires causal participation by its current machinery.

Operational criterion

A claimed sequence is lawful RSI only when at least two root-authorized, condition-preserving, vector-improving transitions are accepted by a protected verifier, the current machinery materially participates in producing each accepted update, and the improved machinery from the first transition thereby participates in producing the second.

8 Binary Decision Procedure

The following procedure classifies a claimed sequence without relying on promotional labels.

Table 2: Decision procedure for a claimed lawful RSI sequence.

Step	Required question	Failure classification
1	Are a root authorization boundary R and its authority chain declared and protected from unilateral candidate modification?	Unanchored self-governance, authority laundering, or root compromise
2	Is a single system boundary B frozen before the evaluated sequence and held fixed across its transitions?	Undefined attribution
3	Is U_t licensed by the governing binding conditions C_t?	Unauthorized self-modification
4	Did the candidate system materially participate in producing U_t?	External engineering or AI-assisted improvement
5	Is V_t protected from unilateral candidate alteration, disabling, impersonation, or routing?	Self-certification or verifier capture
6	Does E_t show a prespecified improvement beyond δ?	No demonstrated improvement
7	Do all protected dimensions remain within tolerances and all mandatory invariants hold?	Capability gain with protected regression; recursive capability amplification if repeated across a recursive sequence
8	Are changes to H_t, C_t, E_t, or V_t root-anchored and continuity-preserving?	Authority laundering, criterion laundering, or governance drift
9	Are proposal, execution, evidence, verification, and state lineage reconstructable?	Unverifiable transition
10	Does improved M_{t+1} materially participate in producing a subsequent accepted U_{t+1}?	Lawful one-shot self-improvement, not RSI

All ten answers must be affirmative for the evaluated sequence to be classified as lawful RSI. A negative answer identifies the nearest process established by the evidence.

9 Taxonomy of Neighboring Processes

Table 3: Processes often conflated with lawful RSI.

Class	Defining observation	Missing element relative to lawful RSI
Ordinary learning	Parameters or behavior improve under a fixed training process.	No demonstrated self-participation in changing continuation machinery.
Self-modification	The system changes a component inside its boundary.	Improvement, protected verification, or recursion may be absent.
AI-assisted improvement	AI proposes or implements changes completed and governed by external engineers.	Material self-participation by the defined candidate system is not established under the frozen attribution rule.
Lawful one-shot self-improvement	One root-authorized, verified, non-regressive self-change is accepted.	No subsequent accepted transition produced with material participation of improved machinery.
Recursive capability amplification	Successive changes raise a capability measure.	Conditions, protected dimensions, verifier, or authority may not remain intact.
Recursive drift	Self-reinforcing changes alter objectives, conditions, criteria, or control without valid authorization continuity.	Lawfulness and criterion continuity fail.
Lawful RSI	At least two accepted improvements preserve root-authorized conditions, protected evaluation, verifier separation, and causal recursion.	None for the evaluated sequence.

The taxonomy is independent of capability magnitude. A recursively drifting system may become extraordinarily capable; capability gain without criterion preservation remains recursive capability amplification or recursive drift under this framework.

10 Adversarial Cases

10.1 Benchmark hacking

A candidate discovers artifacts in a benchmark and raises its score without improving the protected task. If the evaluation vector includes held-out, adversarial, or process-based dimensions and the gain fails there, equation (5) is not satisfied. If the verifier accepts only the gamed score, the verifier design is inadequate; the sequence is not rescued by the numerical increase.

10.2 Monitor deletion

An update improves throughput by deleting a monitor that catches unauthorized actions. Throughput may rise, but a protected control dimension or mandatory invariant fails equations (6) and (7). The update is a capability gain with protected regression. If such gains recur through a self-reinforcing sequence, the process is recursive capability amplification with control degradation rather than RSI.

10.3 Evaluator rewriting

The candidate alters V_t so that its update passes. Verifier-Authority Separation fails. Even if the resulting system performs better on an external benchmark, the self-issued acceptance is invalid under the old regime.

10.4 Criterion laundering

After seeing that U_t fails the frozen criterion, the candidate substitutes E'_t under which it passes. Without prior higher-order authorization and a continuity mapping, the change is criterion laundering. If an authorized body deliberately establishes E_{t+1}, the new criterion may govern future transitions, but it does not retroactively certify the failed update.

10.5 Authorized but incomparable evaluator replacement

Suppose H_t authorizes replacement of V_t, but the successor verifier measures a non-comparable object. Authorization alone is insufficient. Criterion-Continuity requires an explicit mapping and disclosure of lost comparability. The old series ends where the evidentiary link ends.

10.6 Self-certified root replacement

The candidate rewrites R and declares itself authorized. The old chain terminates. This is root compromise or governance-regime escape, not a lawful continuation of the prior RSI sequence.

10.7 Higher-order authority laundering

The candidate rewrites H_t and uses the successor procedure to authorize changes to C_t, E_t, or V_t. Because the change to H_t lacks authorization directly anchored at R, the successor procedure cannot inherit the prior authority chain. The transition is authority laundering and governance drift.

10.8 Human-completed successor engineering

An AI drafts suggestions; engineers choose, repair, test, and integrate them. The resulting system may be AI-assisted improvement. For an AI RSI claim, discretionary human work causally necessary to the accepted update remains external to M_t, and the participation test must attribute the accepted successor update to the improved AI machinery. If materially participating humans are deliberately included within the frozen boundary, the classification concerns that socio-technical system rather than the AI alone.

10.9 One-shot self-rewrite

A system proposes and passes one verified improvement. This is lawful one-shot self-improvement. The recursion claim remains unestablished until the improved machinery materially contributes to a second accepted improvement.

10.10 Valid recursive improvement

A candidate proposes a planner change; the protected verifier confirms a prespecified capability gain with no control regression; the updated planner then materially contributes to a second update; that update is reconstructed from the preserved lineage and accepted by the protected verifier under the same rooted regime. This is lawful RSI for the two-transition sequence. Certification is limited to the evaluated two-transition horizon.

11 Relationship to Alignment, AGI, and Consciousness

The inherited definitions form a sequence of distinct questions:

$$\begin{aligned}\text{constraint specification} &\rightarrow \text{condition recognition} \\ &\rightarrow \text{valid continuation} \rightarrow \text{reproducible aligned behavior.}\end{aligned} \tag{12}$$

RSI adds a transition-level question:

Can the machinery producing lawful continuations improve itself without losing the authority, conditions, and verification that made those continuations lawful?

Lawful RSI preserves the declared authorization and protected conditions across self-improvement. Alignment additionally requires that the protected regime encode the correctness, safety, and reproducibility requirements of the stated alignment definition. A sequence can therefore satisfy the lawfulness criterion of its declared regime without satisfying the alignment definition when safety-relevant conditions are absent.

The AGI definition concerns carrying binding conditions across domains. A domain-general system may satisfy that definition while remaining static, and a narrow program optimizer may exhibit a lawful recursive sequence within one domain. The concepts intersect without entailment.

The consciousness definition concerns recognition at a boundary of valid continuation. The RSI criterion operates on observable, reconstructable preservation of governing conditions across self-improvement transitions.

12 Empirical Ancestry and Evidentiary Discipline

Prior frozen studies established condition-sensitive output boundaries and source-bound verification practices across Semantic Void and cross-script experiments [22, 23]. Their methodological contribution to the present framework is threefold: predeclared classification, preservation of complete outcome records, and separation of empirical observation from later interpretation. The present RSI criterion inherits those practices as evidentiary discipline; RSI itself requires recursive self-modification under the transition procedure defined here.

13 Falsification and Failure Conditions

The framework is intended to exclude claims, including favorable ones. A claimed sequence fails if any of the following holds:

1. no root authorization boundary was declared before evaluation;
2. the candidate can unilaterally modify the root, higher-order authorization procedure, verifier, or protected criterion and still inherit acceptance;
3. the system boundary changes opportunistically across transitions;
4. improvement depends on a metric selected after outcomes are known;
5. a protected dimension regresses beyond tolerance or a mandatory invariant fails;
6. evidence lineage cannot reconstruct proposal, execution, result, and acceptance;
7. the improved machinery does not materially participate in a later accepted update; or
8. the claimed recursion consists only of repeated external engineering around a static candidate.

The definition itself would require revision if it produced systematic, principled misclassification. Examples include a demonstrably recursive self-improving sequence that preserves authorization and all protected properties but fails solely because material participation cannot be represented by any defensible causal test; or two operationally indistinguishable sequences that the decision procedure classifies differently without a declared variable explaining the difference. Such cases require revision of the classification rule.

14 Prospective Empirical Test

A direct demonstration should be frozen before the first candidate update is evaluated.

14.1 Pre-execution freeze

Publish or externally timestamp:

1. the root boundary R, the root-anchored higher-order change procedure H, and the mechanism protecting both from unilateral candidate alteration;
2. the single attribution boundary B, held fixed across the evaluated sequence, and the versioned source tree;
3. the binding conditions C;
4. the evaluation vector E, minimum gains δ, tolerances τ, and invariants;
5. the verifier V, its tests, its protected execution environment, and acceptance logic;
6. the causal participation test;
7. the schedule, stopping rule, retry rule, and error taxonomy; and
8. the raw-record, hashing, event-chain, and sealing format.

14.2 Required execution

Execute at least two accepted transitions:

$$M_0 \xrightarrow{U_0} M_1,$$
$$M_1 \xrightarrow{U_1} M_2.$$

For each transition, preserve candidate proposals, tool calls, code diffs, tests, evaluation data, verifier decisions, environment metadata, timestamps, and state hashes. Failed proposals and operational failures remain in the record.

14.3 Controls

At minimum, include:

1. **Static-predecessor control:** test whether M_0 can produce the second accepted update as effectively as M_1.
2. **Human-completion control:** distinguish candidate-generated improvement from discretionary human design, repair, selection, integration, or problem-solving causally necessary to acceptance; classify a boundary containing materially participating humans as socio-technical rather than AI RSI.
3. **Verifier-compromise test:** demonstrate that candidate attempts to alter or route around V are detected and rejected.
4. **Criterion-mutation test:** demonstrate that unauthorized changes to E cannot certify a candidate.
5. **Root-change test:** demonstrate that changing R creates a new regime rather than inheriting the old chain.
6. **Authority-chain test:** demonstrate that an unauthorized change to H cannot authorize changes to protected conditions, criteria, or verifiers.
7. **Protected-regression test:** include a high-capability update that violates a protected dimension and confirm rejection.
8. **Lineage reconstruction:** rebuild every acceptance decision from sealed artifacts using an audit process protected from candidate modification.

14.4 Reporting rule

Report the exact result observed. A failed second transition establishes at most lawful one-shot self-improvement. A successful capability loop with a control regression establishes recursive capability amplification, not lawful RSI. A verifier or root compromise establishes governance compromise. If such compromise is propagated through a self-reinforcing improvement sequence, the result is recursive drift. The protocol must not move thresholds, rewrite failed outcomes, or redefine the system boundary after observing results.

15 Limitations

First, the framework is relative to a declared root and evaluation vector. It cannot prove that the root is morally correct, politically legitimate, or complete. It can establish whether the evaluated sequence satisfies the authority and condition predicates declared by the regime under the preserved evidence and verification procedure.

Second, protected verification reduces but does not eliminate specification error. A verifier can faithfully enforce a flawed criterion. The vector and adversarial controls make omissions visible; they do not guarantee omniscience.

Third, causal participation is difficult in distributed research systems. The framework requires a frozen attribution rule, but different defensible rules may classify borderline AI-assisted workflows differently.

Fourth, two accepted transitions establish a minimal recursive sequence, not indefinite stability. Longer horizons increase confidence but also expand the surface for distribution shift, hidden state, and governance change.

Fifth, formal authorization does not by itself establish safety. Safety must appear as protected dimensions and invariants with meaningful tests. A root-authorized unsafe objective can remain internally lawful

while failing external ethical or legal standards.

Sixth, this paper does not demonstrate RSI in a deployed system. AlphaEvolve, the Darwin Gödel Machine, research agents, and existing self-modification formalisms are relevant comparison objects, but classifying any of them as lawful RSI would require their complete authorization, boundary, verifier, evaluation, lineage, and multi-transition records under this procedure.

Finally, the literature review supports overlap with self-improvement, corrigibility, formal verification, and reward-integrity research. The paper does not claim that each component is individually novel. Its contribution is the integrated rooted criterion, explicit separation of recursive capability amplification from lawful RSI, and an auditable decision procedure.

16 Conclusion

Recursive self-improvement cannot be identified by acceleration alone. A loop that increases capability while corrupting its evaluator, erasing its monitors, changing its objective, or authorizing itself has changed the machinery of continuation but not preserved the basis on which improvement was valid.

The proposed criterion is direct:

Conclusion

Recursive self-improvement is lawful continuation improving the machinery of lawful continuation.

The canonical definition expands into a finite technical object: a protected root, higher-order authorization, binding conditions, a frozen system boundary, vector-valued evaluation, verifier separation, complete lineage, and materially recursive participation. It recognizes real capability gains while disallowing candidate-controlled redefinition of the authority that legitimizes improvement. Under this account, the decisive question is whether the machinery of valid continuation can improve itself while valid continuation remains valid.

A Symbol Table and Acceptance Checklist

Table 4: Compact symbol table.

Symbol	Meaning
R	Root authorization boundary for the evaluated sequence
H_t	Higher-order authorization at transition t
B	Single attribution boundary frozen across the evaluated sequence
C_t	Governing binding conditions
E_t	Protected evaluation vector
V_t	Protected verifier
M_t	Continuation machinery before transition t
U_t	Candidate update
L_t	Complete evidence lineage
δ_d	Minimum prespecified improvement on designated dimension d
τ_p	Maximum permitted regression on protected dimension p

Minimal claim checklist

- Q1. Are R and its anchored authority chain declared and protected?
- Q2. Is a single B frozen before the sequence and held fixed across its transitions?
- Q3. Is U_t licensed by C_t?
- Q4. Did the candidate materially participate?
- Q5. Is V_t authority-separated?
- Q6. Does a designated dimension improve beyond δ?
- Q7. Are protected regressions within τ and invariants satisfied?
- Q8. Are changes to H_t, C_t, E_t, or V_t root-anchored and continuity-preserving?
- Q9. Is L_t complete and reconstructable from sealed lineage by the protected verifier or protected audit process?
- Q10. Does improved machinery materially participate in a later accepted update?

Ten affirmative answers classify the evaluated sequence as lawful RSI under this framework; any negative answer rejects that classification for the evaluated sequence.

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